

Performance

A hands-on course in building agents by building their harness

Lesson 9

A run that finishes can still cost too much

- Lesson 8 made the agent keep going when something fails
- But ten minutes and ten dollars is a run you think twice about starting
- Over a long task two things add up: **the time you wait** and **the tokens you pay for**

Time and tokens come from the same place

Primitive	Where the time and tokens go
Context	what the request holds, resent on every pass
Model interface	waiting on the model to read and write
Output	waiting on commands to finish

A production layer, not a new primitive. Control flow, input and the sandbox are left as they are.

The cost of a task is everything resent

- Every pass of the loop sends the **whole** request again: tools, instructions, every message
- Twenty steps means reading the same opening twenty times
- **Prompt caching:** the request is a prefix, read in a fixed order
- Mark where the remembering ends with `cache_control`; a call that starts with the same tokens only reads the new end
- A cache read costs a tenth of normal input; a write, a quarter more; it lasts five minutes since last use

A cache has three catches

- **The match has to be exact.** One changed character early and everything after it is new. That's why Lesson 4's prompt has the date, not the time
- **There's a minimum.** 512 to 4,096 tokens depending on the model; quark's system prompt is about nine thousand tokens
- **It belongs to one model.** Switch models and you start again

The system prompt has carried a mark since Lesson 4. New here: a second mark on the **last message**.

Send less, wait less

- **Trim:** cap what one tool result may add, keeping the start and the end
- **Stream:** the reply arrives word by word, starting after half a second, not all at once after eight
- **Small jobs to the small model:** compaction always goes to a faster, cheaper one
- **Run together:** several tool requests in one reply run at the same time
- Results still go back in order, each with its id; the guard's questions come first, one at a time

`call()` streams, model by model (abridged)

```
MODELS, FAST = ["claude-sonnet-5-5", "claude-opus-5-5"], ["claude-haiku-4-5"]
def call(models=MODELS, live=False, **request):
    for model in models:
        try:
            with client.messages.stream(model=model, **request) as stream:
                shown = False
                for event in stream:
                    if ESC.is_set(): break
                    if live and ...text_delta": print(event.delta.text, end="", flush=True); shown = True
                if shown: print()
            return stream.current_message_snapshot if ESC.is_set() else stream.get_final_message()
```

Compaction asks for the small model: `call(models=FAST, ...)` .

Context: trim, and cache the end (abridged)

```
MAX_RESULT = 20_000
def trim(text):
    if len(text) <= MAX_RESULT: return text
    return text[:MAX_RESULT // 2] + f"\n[... {len(text) - MAX_RESULT} characters cut ...]\n" + ...

def cached(working_memory):
    last = working_memory[-1]
    blocks = [{"type": "text", "text": last["content"]} if isinstance(last["content"], str) else ...
    blocks[-1] = {**blocks[-1], "cache_control": {"type": "ephemeral"}}
    return working_memory[:-1] + [{"role": last["role"], "content": blocks}]
```

The main call sends the marked copy, streamed live:

```
with listening():
    response = call(live=True, max_tokens=16384, ..., messages=cached(working_memory))
```


The stream makes ESC immediate (abridged)

- `call()` checks ESC at every piece that arrives, and stops reading the moment you press it, mid-thought or mid-sentence
- It hands back what had arrived so far: `stream.current_message_snapshot`
- Text that was said, up to where you stopped it, stays in working memory and the episode
- Thinking that wasn't finished is dropped: the API only takes back thinking it signed

```
output = [b for b in response.content
           if (b.type != "text" or b.text) and (b.type != "thinking" or b.signature)]
if output: add(working_memory, {"role": "assistant", "content": output})
```

Output: one command becomes a function (abridged)

```
def execute(cmd):
    start = time.time()
    doing = subprocess.Popen(["docker", "exec", box, "timeout", ..., cmd], stdout=subprocess.PIPE, ...)
    while True:
        try: done = subprocess.CompletedProcess(doing.args, 0, doing.communicate(timeout=0.1)[0]); break
        except subprocess.TimeoutExpired:
            if ESC.is_set(): subprocess.run(["docker", "exec", box, "sh", "-c", "kill -9 -1"], ...)
    done.returncode = doing.returncode
    if ESC.is_set(): done.stdout += "\n[your doing stopped before done]"
    elif done.returncode == 137: done.stdout += f"\n(killed: ...)"
    trace(event="tool", cmd=cmd, seconds=..., exit=done.returncode, chars=len(done.stdout))
    return trim(done.stdout) or f"(exit {done.returncode})"
```

One line of the prompt changes too: commands that don't depend on each other can go in the same response, *"they run at the same time."*

Decide each request first, in order (abridged)

```
refused, pending = {}, {}  
for block in output:  
    if block.type == "tool_use":  
        cmd = block.input.get("cmd")  
        print(f"$ {cmd}")  
        if not cmd or (response.stop_reason == "max_tokens" and block is output[-1]):  
            refused[block.id] = "your request was cut off at the token limit, so it was not run. ..."  
        elif (no := guard(cmd)):  
            trace(event="refused", cmd=cmd, why=no)  
            refused[block.id] = no  
        else:  
            pending[block.id] = cmd
```

Then run everything allowed, at once (abridged)

```
with listening(), ThreadPoolExecutor() as pool:
    outputs = dict(zip(pending, pool.map(execute, pending.values())))

input = []
for block in output:
    if block.type == "tool_use":
        text = refused.get(block.id) or outputs[block.id]
        print(f"[{text}]" if block.id in refused else text)
        input.append({"type": "tool_result", "tool_use_id": block.id, "content": text, ...})
```

Three seconds of waiting, not nine

Three `sleep 3` commands that don't depend on each other. This lesson's `quark.py`:

```
$ sleep 3; echo lint ok
allow `sleep 3; echo lint ok`? [y/N] $ sleep 3; echo types ok
allow `sleep 3; echo types ok`? [y/N] $ sleep 3; echo tests ok
allow `sleep 3; echo tests ok`? [y/N] lint ok

types ok

tests ok

Lint, types, and tests all passed.
[wall: 8.0s]
```

Lesson 8's `quark.py` ran them one after another: `[wall: 14.6s]` .

The trace shows it, and the cache too

Lesson 8:

```
{"event": "model", "seconds": 2.35, "input_tokens": 145, "cache_read": 0, "cache_write": 8438}  
{"event": "tool", "seconds": 3.09, "cmd": "sleep 3; echo lint ok"}  
{"event": "tool", "seconds": 3.11, "cmd": "sleep 3; echo types ok"}  
{"event": "tool", "seconds": 3.11, "cmd": "sleep 3; echo tests ok"}  
{"event": "model", "seconds": 1.12, "input_tokens": 407, "cache_read": 8438, "cache_write": 0}
```

This lesson:

```
{"event": "model", "seconds": 1.73, "input_tokens": 4, "cache_read": 0, "cache_write": 9130}  
{"event": "tool", "seconds": 3.12, "cmd": "sleep 3; echo lint ok"}  
{"event": "tool", "seconds": 3.12, "cmd": "sleep 3; echo tests ok"}  
{"event": "tool", "seconds": 3.13, "cmd": "sleep 3; echo types ok"}  
{"event": "model", "seconds": 1.24, "input_tokens": 2, "cache_read": 9130, "cache_write": 264}
```

Each call pays full price for a handful of tokens

A three-step task, one command per step:

```
{"seconds":1.9,"input_tokens":4,"output_tokens":69,"cache_read":0,"cache_write":9118}  
{"seconds":1.74,"input_tokens":2,"output_tokens":75,"cache_read":9118,"cache_write":155}  
{"seconds":2.19,"input_tokens":2,"output_tokens":105,"cache_read":9273,"cache_write":125}
```

- `input_tokens` is what was read at full price: 4, then 2, then 2
- The first call wrote about 9,100 tokens to the cache, mostly the system prompt
- Each later call wrote only what was new: 155 and 125 tokens

A trimmed result is never resent

`cat docs/the-model.md`, a file over 25,000 characters:

```
{"event":"model","seconds":1.54,"input_tokens":4,"cache_read":0,"cache_write":9088}  
{"event":"tool","seconds":0.11,"chars":25435}  
{"event":"model","seconds":2.16,"input_tokens":2,"cache_read":9088,"cache_write":6533}
```

- `chars` is the length before trimming: 25,435
- The second call cached 6,533 tokens: the trimmed result
- The 5,435 characters cut would otherwise be resent on every later call

One ESC stops both commands (shortened)

```
$ for i in 1 2 3 4 5 6 7 8 9; do echo a$i; sleep 1; done
allow `for i in 1 2 3 4 5 6 7 8 9; do echo a$i; sleep 1; done`? [y/N] y
$ for i in 1 2 3 4 5 6 7 8 9; do echo b$i; sleep 1; done
allow `for i in 1 2 3 4 5 6 7 8 9; do echo b$i; sleep 1; done`? [y/N] y
a1
...
a5

[your doing stopped before done]
b1
...
b5

[your doing stopped before done]
```

Then the model: *"You interrupted me, so I stopped both commands."*

What else performance can be

- **What gets cached:** up to four marks; a one-hour lifetime for agents that wait on people
- **What goes in:** clear old tool results; let the model fetch what it needs
- **How much comes back:** "be brief" does more than `max_tokens` ; lower thinking effort
- **How many at once:** cap parallel tools; the batch interface at about half the price
- **As a product:** the API's own settings; [LiteLLM](#) and [OpenRouter](#) choose models per request

performance.py: a small model routes

Lesson 3's loop, async, measuring itself. One call to a small model routes each task:

```
ROUTER = "claude-haiku-4-5"
TIERS = {                                     # tier -> (model, effort)
    "quick": ("claude-haiku-4-5", None),
    "standard": ("claude-sonnet-5-5", "medium"),
    "deep": ("claude-opus-5-5", "high"),
}
```

- Every call prints time to first output and how input splits: new, from cache, written
- Its prompt holds the project's README and docs/, so there's something to cache

Warm cache, cheaper bill; a guess, wrong

```
[done in 2 steps, 1.7s, 50% of input tokens read from the cache]  
[done in 2 steps, 1.9s, 100% of input tokens read from the cache]
```

Same task twice: 50% of input from the cache cold, 100% warm. The saving was in the bill, not the clock.

```
[routed to deep: claude-opus-5-5]  
[3 commands: 3.0s together, 9.0s one after another]
```

Three `echo` s sent to the deep tier: more model than they need.

The rule

Don't resend what hasn't changed, don't send what isn't needed, and don't wait for things one at a time that could be waiting together.

Performance changes how, never what

- **Control flow:** the same loop, the same stops, the same number of calls
- **Input:** the person's task is read the way it always was
- **Context** still decides what the request holds, only marked for reuse
- **Model interface** still sends a request and gets a reply, only streamed
- **Output** still shows the reply and runs what it asks, only sooner and together

Faster and cheaper, but as good?

- A trimmed result might have cut the line that mattered
- A summary from the small model might have lost a fact
- The router might send a hard task to a weak model
- Parallel commands can quietly depend on each other
- The trace shows a faster, cheaper run, and not whether it did the job

Next: Evaluation

Lesson 10